

Developing metacognitive knowledge for effective AI-supported study behaviors in undergraduate biology: recommendations for instructors

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ABSTRACT The use of effective study behaviors is essential for student academic performance and persistence. Currently, it is reported that college students use generative artificial intelligence (AI) to help them prepare for courses, suggesting they hold metacognitive knowledge of AI as a study tool. However, there is no set of recommendations to develop metacognitive knowledge of effective AI-supported study behaviors for students enrolled in introductory biology courses. In this essay, we draw on existing literature on study strategies, metacognition, and emerging studies on AI use in higher education to consider how students use generative AI to support their learning and how incomplete metacognitive knowledge of AI use can result in continued use of ineffective study strategies that negatively influence academic performance. We then offer recommendations to support the development of this knowledge in undergraduate courses. As generative AI continues to become a mainstay within higher education, we hope this essay equips instructors with the knowledge to promote student success in biology coursework.

KEYWORDS generative AI, metacognition, study strategies, AI-supported learning, undergraduate biology

Large-enrollment introductory biology courses often present significant challenges for students that influence their decisions to persist in science, technology, engineering, and mathematics (STEM) fields. One significant challenge that students encounter is low academic performance and the use of ineffective study behaviors, such as rereading the textbook or rewriting notes word for word (1). Effective study behaviors are defined as behaviors that facilitate long-term retention (2–5). These behaviors encourage the retrieval of information and include self-testing (e.g., using flashcards and completing old exams or quizzes), summarizing (e.g., creating a study guide), self-explanation (e.g., making diagrams and explaining concepts to others), interleaving, and spaced study (6). Previous studies demonstrate that while the use of effective study behaviors is associated with higher academic performance in introductory biology courses, students frequently report using a mix of effective and ineffective study strategies (i.e., strategies that promote fluency and rote memorization, such as repeated rereading or rewriting notes) (2, 6, 7). These studies suggest that students hold some metacognitive knowledge regarding how they should study for their courses. Specifically, students can name the study strategies that are available for them to use (i.e., declarative metacognitive knowledge) but may struggle in the implementation of specific study strategies (i.e., procedural and conditional knowledge) and evaluation of their effectiveness. Collectively, these gaps in metacognitive knowledge and regulation may result in students continuously using ineffective study strategies despite their negative impacts on course performance.

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Given the role of study behaviors on academic performance, multiple studies have explored the benefits of implementing co-curricular study skill workshops and in-class instruction on student academic performance (8–12). Similarly, other studies have advocated for the direct instruction of study skills through course materials (e.g., syllabi) (13). However, Rea and colleagues (14) found that despite knowing effective study strategies (i.e., they can name what they are), students often chose not to implement them due to time constraints and low self-efficacy, suggesting that students have limited procedural and conditional knowledge of these strategies.

In addition to the curricular support that helps develop student study skills, students also make use of current technology, such as generative artificial intelligence (AI), to support their learning independently. Generative AI has garnered much attention across higher education over the past few years (15). While current studies have characterized the negative impacts of excessive AI use by students (e.g., procrastination, memory loss, and declines in academic performance) (15–18), AI use has also been demonstrated to improve student understanding of content across multiple disciplines, including biology (19–21). Furthermore, studies consistently demonstrate that undergraduate students use AI as a time-saving measure to support their learning (e.g., using AI-generated study materials) (15, 22, 23). In the context of study strategies, students' use of AI is influenced by their metacognitive knowledge of study strategies and their metacognitive knowledge of AI use generally (24–26). As a result, student AI use can either reinforce the use of effective study strategies for students who have well-developed metacognitive knowledge or encourage the continued use of ineffective strategies for students who do not (27). In this essay, we use the metacognition framework to propose that student AI use is a form of metacognitive knowledge that influences how students study. We also characterize challenges students may encounter when using generative AI independently and propose approaches that instructors can use to address them.

THEORETICAL FRAMEWORK

In this essay, we conceptualize student use of generative AI tools as a form of study strategy-related metacognitive knowledge. Metacognition has been identified as a key factor in facilitating academic success in undergraduate biology and STEM broadly, as it encourages the use of evidence-based study strategies and encourages student self-evaluation (28–31). Therefore, we draw from the metacognition framework to highlight factors that support the effective implementation of AI tools for studying. Briefly, metacognition is defined as a learner's awareness of their own learning and cognitive processes and consists of two components: metacognitive knowledge and metacognitive regulation (32). Metacognitive knowledge refers to what students know about their cognitive processes and specific learning strategies (declarative knowledge), their knowledge regarding the implementation of those strategies (procedural knowledge), and their knowledge of when to use those strategies and why (conditional knowledge) (33). In undergraduate biology contexts, a student with strong metacognitive knowledge regarding the use of generative AI as a study strategy might know that self-testing is effective for long-term retention and that AI can be used as a study tool (declarative knowledge). The same student may also know that self-testing involves taking a practice test and that generative AI can be used to generate practice tests (procedural knowledge) and that this strategy is especially useful for preparing for application-based exams (conditional knowledge). Metacognitive regulation refers to actions students take to learn and involves selecting specific study strategies to use (planning), the implementation and active assessment of those strategies (monitoring), and determining the effectiveness of those individual strategies (evaluation) (29, 33). This aspect of metacognition is especially important as it allows students to accurately gauge what they do and do not know (Fig. 1).

Studies demonstrate that students use generative AI as a study tool and perceive it to be an effective way to support their preparation for exams (36–41) (declarative knowledge). However, this metacognitive knowledge is incomplete. While students can name

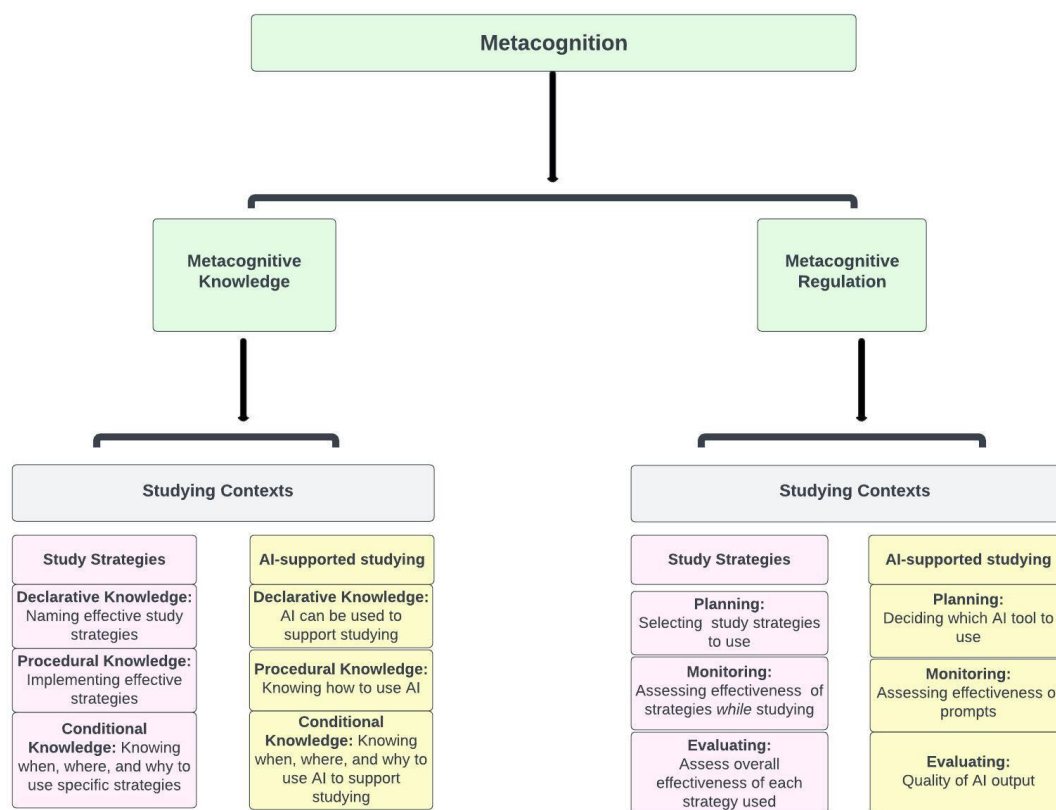


FIG 1 Conceptual illustration of how metacognitive knowledge and regulation operate across two studying contexts: study strategies and AI-supported studying. AI-related components represent an extension of the metacognitive framework described by Xiao et al. (34) and Schraw and Dennison (35).

AI as a tool they can use (declarative knowledge), student variation in how they use it demonstrates a lack of *procedural* metacognitive knowledge in how to leverage AI to support the use of effective study strategies (e.g., self-testing and self-explanation) (42). For example, some students may use AI as a way to engage in self-testing. These students will upload their notes from lectures into a generative AI tool (e.g., Chat Generative Pre-Trained Transformer [GPT]) and command it to generate questions from those notes. Students will then use these questions to quiz themselves. Alternatively, other students may upload their notes and command generative AI to generate a summary that they reread repeatedly, further supporting the use of ineffective strategies. This variation in procedural metacognitive knowledge may be accompanied by differences in *conditional* metacognitive knowledge, where students may not know when to implement certain strategies and why. For example, if a student understands that self-testing is an effective strategy to prepare for exams that are application based, that understanding will reflect in how they use generative AI to study.

Recent studies demonstrate that students use AI to support their learning, which indicates that students have their own metacognitive knowledge of AI used for the purpose of studying (39). While students know that AI can be used to assist with studying, they have varying levels of knowledge regarding its implementation, which could influence which study strategies students choose to use (42, 43). Undergraduate students enrolled in biology also display uneven metacognitive abilities, particularly in planning and evaluating their study strategies (2, 33, 44, 45). Furthermore, research shows that students know of strategies but lack the procedural or conditional knowledge to apply them effectively (45, 46). We propose that, rather than viewing use of generative AI as an overall threat to academic integrity, the metacognitive framework can (i) help instructors better understand how students' use of AI reflects metacognitive knowledge

of study approaches and (ii) provide instructors with intervention targets to support effective use of AI while also encouraging the use of effective study behaviors.

SCOPE OF THE ESSAY

Current literature focuses on the use of generative AI for assessment purposes in undergraduate biology. Here, we leverage the metacognition framework to consider how students use AI to support their learning within biology. Specifically, this essay discusses the following:

1. Student metacognitive knowledge of generative AI and how they use it to support their learning.
2. Recommendations that we can provide to develop metacognitive knowledge of generative AI and encourage the use of effective study strategies.

POSITIONALITY STATEMENT

This essay is grounded in the shared experiences among the authors regarding the use of generative AI in undergraduate biology contexts. All the authors are students and instructors at a large, research-intensive, public university in the southeast. Four of the authors (V.A., M.A., P.M., K.R., and F.B.) are students who have enrolled in large, enrollment introductory biology classes as students and/or have served as teaching/learning assistants in these courses, where they have used generative AI for study purposes themselves and/or interacted with peers who use AI for this purpose. Additionally, they have encountered courses that either encourage the use of generative AI or expressly forbid it. One author (S.N.E.) is a junior faculty member who teaches upper-level biology courses where they explicitly discourage students from using AI to complete course assignments but encourage students to use generative AI for the purposes of studying, if they so choose. In addition to identities related to faculty membership and students, the authors also hold a range of other identities across race/ethnicity, gender, neurodiversity, international status, and socioeconomic status. Collectively, these identities and experiences inform their perspectives related to education and the use of generative AI.

How might undergraduate biology students use generative AI?

In undergraduate biology, current studies have centered on the use of AI for assessment (47–49). To date, there are, to our knowledge, no studies that explore how students enrolled in undergraduate biology coursework leverage the use of generative AI to support their learning, highlighting a potential area for future research. Therefore, this section synthesizes current research that characterizes student interactions with generative AI tools across higher education to consider how undergraduate biology students use AI to support their learning.

With the launch of ChatGPT-4, generative AI is able to analyze images, documents, and diagrams in both professional and academic contexts, providing more opportunities for students to leverage generative AI for exam preparation (50). While instructors may suspect that students are using AI to cheat on assignments, the literature indicates that, across disciplines, undergraduate students are increasingly using AI as a study tool, with students using AI for quick clarifications related to course content (e.g., “What are the stages of mitosis?” and “What is X-chromosome inactivation?”) and summaries of complex concepts, and generating practice materials such as flashcards or quiz questions (51). Students also report using AI-based personalized tutoring systems that provide immediate feedback, adapt to student learning preferences, and align with their needs. These platforms have led to an increase in motivation and minimized feelings of overwhelm, especially for STEM students (27, 51). A study analyzing the impact of AI use reveals that these tools are allowing students to implement more effective study strategies into their routines, resulting in slightly higher GPAs and reduced study time (51). This suggests that AI may help students simplify and organize study routines

without reducing performance (48). Perhaps unsurprisingly, AI use for the purpose of studying is pervasive. While some students may enthusiastically use AI to prepare for exams, others may avoid using AI-powered study tools such as ChatGPT due to negative perceptions of these tools (52–54). These perceptions may stem from additional ethical concerns, including data privacy concerns, inauthenticity concerns, content bias, intellectual property issues, and impact on student independence (55). However, these students may have still unknowingly come into contact with these tools through the use of Quizlet or other study-based apps that are frequently used to promote the use of effective study strategies, such as self-quizzing (27).

Despite the potential role of AI in promoting the use of effective study behaviors, there are also substantial risks associated with its use as a study aid that may undermine the learning process that it is intended to support. Specifically, AI is not designed for the purpose of education and generates responses based on predictions of word sequences provided in training data rather than actual reasoning or comprehension (56). As a result, AI may produce responses that sound coherent but lack intellectual grounding. Students may lack the content knowledge necessary to critically evaluate what AI presents to them and, as a result, often display blind trust in AI and mistake output generated as evidence of validity, suggesting limited metacognitive engagement. Consistent with this, in a study of over 300 student ChatGPT conversations, Zheng and colleagues (57) found that many students either relied too heavily on AI or dismissed it altogether, with both patterns resulting in lower scores on post-task assessments of conceptual understanding, indicating limited deeper engagement with the content. Abdelhalim (42) demonstrated that this variation in AI use reflects underlying differences in student metacognition. Students with high metacognitive awareness of AI (i.e., students with high metacognitive knowledge and metacognitive regulation) use AI as a tool to support their learning and critically evaluate its output. Conversely, students with lower metacognitive awareness used AI to obtain quick answers. Similarly, Xiao et al. (34) found that students' metacognitive knowledge of AI tools (e.g., when, where, and how to use them) developed based on student engagement with and evaluations of those tools. Consistent with these findings, Lodge et al. (58) and Bearman et al. (59) posit that effective AI use requires metacognitive skills that allow students to not only assess the quality of AI-generated output but also identify areas of conceptual weakness.

AI use may also encourage cognitive offloading, where students limit the amount of cognitive effort needed to complete a task (53, 60). As a result, student use of AI may support further use of ineffective study strategies (e.g., rereading). For example, due to time pressure, lack of confidence, or just an easier way to complete assignments, AI allows students easy access to explanations, encourages prioritization of fast solutions, promotes surface-level engagement, and gives students the ability to pass off AI-generated work as their own while earning better grades (52, 53). Micabalo and colleagues (27) found that overreliance on aid from AI leads to passive learning (e.g., rereading AI-generated summaries) and does not require students to engage in critical thinking or problem solving. Taken together, current research suggests that while generative AI may support student learning, its impact is dependent on how students engage with these tools and may reinforce existing study behaviors.

RECOMMENDATIONS TO SUPPORT THE DEVELOPMENT OF COMPLETE METACOGNITIVE KNOWLEDGE REGARDING THE USE OF GENERATIVE AI AS A STUDY STRATEGY

Many of the current studies exploring AI have found that students consistently use it as a study partner, demonstrating metacognitive knowledge of its utility as a facilitator of effective studying (34, 39, 61). However, major confounds associated with AI use for this purpose are the accuracy of AI-generated information and the need for students to regularly evaluate the information presented (34, 59). This indicates that, in addition to holding declarative metacognitive knowledge about AI, students must also develop procedural and conditional metacognitive knowledge about its use, especially as it

relates to its role in supporting student learning. Consistent with these findings, many instructors agree that AI education should be a mandatory requirement for students in today's technology-led society (62).

Explicit instruction has been demonstrated to promote metacognition and the use of effective study strategies, and we recommend that instructors provide explicit instruction regarding AI use for the purpose of studying (63–67). In line with this, metacognitive training modules and exam reflections have been shown to increase self-regulation and critical thinking, as well as increase the use of effective study strategies for students enrolled in undergraduate biology courses (68, 69). Conversely, without direct instruction, students frequently adopt ineffective strategies (e.g., rereading summaries) that do not promote effective learning (70). Therefore, explicit instruction that leverages the metacognition framework, instead of simply being provided with declarative metacognitive knowledge (i.e., being told that AI can be used to study), provides students with information about what AI is, when it is appropriate to use it (conditional knowledge), how AI can be used to support studying (procedural knowledge), and feedback regarding their use of AI for this purpose. Collectively, this information serves to deepen student metacognitive knowledge on AI use (Table 1).

While explicit instruction is beneficial in developing metacognition, given the novelty of AI, we recognize that undergraduate biology instructors possess varied understanding and comfort with AI technologies. Multiple studies show that undergraduate science instructors positively favor the use of AI for teaching biology (71) and generally hold a positive attitude towards integrating AI into teaching (72). However, in using these tools, instructors must carefully review and edit AI-generated responses, implying the need for sufficient knowledge of AI's capabilities and limitations (73, 74). Instructors may have had limited opportunities to develop these AI competencies and may remain hesitant to independently integrate these tools without broader guidance (74, 75). This need for readily accessible guidance is vital as educators play an irreplaceable role in interpreting AI outputs, contextualizing knowledge, and integrating ethical considerations into biological science education (76). It is our hope that recommendations in this essay provide instructors with information regarding the opportunities and challenges of AI use, as well as guidance on how to leverage this tool specifically for the purpose of studying. Instructors who wish to further build upon the foundational knowledge presented in this article are encouraged to reach out to institutional faculty learning centers (e.g., Center for Teaching and Learning), discipline-based education communities, or emerging guidelines on AI use in higher education (e.g., U.S. Department of Labor's Artificial Intelligence Literacy Framework).

We also acknowledge that instructors in introductory biology courses may have structural constraints that limit the implementation of interventions that support AI-based study practices. For example, at a given institution, introductory biology courses may be highly coordinated across sections with shared syllabi, assessments, and

TABLE 1 Recommendations to develop student metacognitive knowledge

Recommendation	What this targets	Why it matters	Example instructional action
Develop declarative knowledge	Awareness of what AI is, its benefits, and limitations	Students have limited knowledge of how to use AI or feel guilty about its use	The instructor introduces different AI tools and discusses their benefits and limitations (e.g., hallucinations)
Develop procedural knowledge	How to use AI effectively to support studying	Students default to summary generation and ineffective study strategies (e.g., rereading)	The instructor models how to input prompts that encourage the use of effective study strategies
Develop conditional knowledge	When and why to use AI	Students misalign study strategies with exam type	The instructor shows how exam format informs strategy
Support metacognitive regulation	Monitoring and evaluating AI output	Students must evaluate the accuracy and usefulness of AI output and adjust their study strategies accordingly	The instructor provides AI-generated responses and asks students to identify errors, assess accuracy, and determine whether the output supports effective studying

course policies. This may restrict individual instructors' flexibility in encouraging AI use. Instructors in these contexts are encouraged to adapt the recommendations presented here within existing course policies and engage in broader departmental discussions regarding the role of AI in student learning. Furthermore, the metacognitive principles discussed in this article can be applied to student study practices generally. Therefore, instructors are able to facilitate metacognitive development and the use of effective study behaviors in contexts where AI use is limited or discouraged.

Finally, we emphasize that we do not advocate for the adoption or rejection of AI in undergraduate biology contexts. Instead, we provide recommendations, rooted in the U.S. Department of Labor's Artificial Intelligence Literacy Framework (77) and best practices described in metacognition and study strategy literature, that are intended to scaffold student use of AI in a way that encourages responsible AI use by developing students' declarative, procedural, and conditional metacognitive knowledge related to generative AI use for studying for students who choose to use AI for this purpose.

Recommendation 1: explicitly develop student declarative metacognitive knowledge about AI use

Students' use of AI to support their study strategies may be influenced by their perception of the tool generally and their knowledge regarding its benefits and limitations. For example, some students may hold negative opinions of AI and, as a result, will not use it to support their exam preparation. For some students, this may serve to negatively impact their academic performance, particularly those students who, due to structural barriers and inequities, have not had opportunities to access resources that encourage the use of effective study strategies (13, 78). Additionally, given the novelty of generative AI, students may not be fully aware of its limitations as a study aid. For example, while generative AI can summarize complex concepts easily, the information generated may not be accurate due to lags in system updates (79–81). Below, we discuss how instructors should use direct instruction to address students' feelings of guilt surrounding AI use and inform them of pitfalls that are also associated with its use to further develop declarative knowledge of generative AI.

Address feelings of guilt about using AI (AI guilt)

The mystery and confusion surrounding artificial intelligence have been shown to serve as a source of anxiety for students, which ultimately diminishes the potential positive outcomes of the technology (82). The specific anxiety is known as "AI guilt" (83). AI guilt is a psychological experience where students feel guilt and moral discomfort surrounding their use of artificial intelligence, which mostly presents as a fear that the use of AI will be perceived negatively (e.g., being perceived as lazy or inauthentic) by others (83). This fear is also accompanied by increased stress and anxiety as well as low self-efficacy (83). In environments that prioritize individual creativity and work ethic, like educational settings, this phenomenon may even be heightened (83). Although a student may be able to complete a task using AI, knowing that they did not do so independently may leave them feeling ultimately unsuccessful and ashamed. Negative emotions, similar to those caused by AI guilt, have been shown to negatively affect metacognitive processes (84). Therefore, AI guilt is a prominent issue plaguing students and may decrease students' willingness to acknowledge the positive impacts of the technology due to ethical dilemmas (82, 85). Not only are students less likely to use the technology, but they are also less likely to recommend the use of AI to their peers (82). To our knowledge, no studies have been conducted examining AI guilt in biology courses, and this is an area where research should be expanded. Providing students with information that allows them to understand how to interact with AI while also acknowledging the social and ethical outcomes of the program (i.e., "AI literacy") has been demonstrated to be an effective tool to combat student AI guilt (86). A study conducted by Brown and colleagues (86) demonstrated that one-third of participants believed university guidance on AI use (e.g., when it is appropriate to use) was unclear, contributing to

student anxiety regarding its use. We recommend that instructors reframe AI in a positive light, emphasizing that, when used appropriately, AI is an effective educational tool. Additionally, instructors can encourage honesty and transparency among their students regarding their use of AI for classwork (83). Instructors can also implement “AI literacy courses” woven into the already established curriculum (87). These courses should cover each of the basic components of AI literacy: understanding the basic architecture of the technology, acknowledging the limitations, comprehending the risks and benefits, and discussing the societal and ethical implications. Through implementation of these practices, instructors may be capable of combating AI guilt and the negative emotions inhibiting the development of student declarative metacognitive knowledge surrounding AI use.

Inform students of pitfalls associated with AI use and demonstrate how to avoid them

We also encourage instructors to consider the veracity of the information that AI produces and to inform students of this pitfall. OpenAI has stated that ChatGPT can produce intelligent-sounding yet completely incorrect answers (i.e., “hallucinations”) (50). Similarly, Tlili et al. (88) reported that even with a consistent topic, ChatGPT generated variable responses based on the prompt that it was given. However, given the novelty and convenience of AI, students are likely to readily accept the responses generated by ChatGPT and may not spend time evaluating the output, which could result in misconceptions regarding course content. A survey distributed to 1,250 college students in the United Kingdom revealed that while a third of respondents reported using generative AI as a personal tutor, they also did not know if and how often generative AI was producing hallucinations (89). Therefore, we recommend that instructors inform students of this characteristic of AI. Instructors who feel comfortable should also demonstrate to students how to input content and lecture notes directly from a course to generate both personalized and reliable study tools. After inputting this information, artificial intelligence can create accurate quizzes in a number of different formats (e.g., multiple choice, true/false, short answer, and matching) that can assess rote memorization or encourage application of course content to real-world scenarios (90).

Recommendation 2: explicitly develop procedural knowledge of AI use for studying

Show students how to effectively use AI to generate active learning study tools and promote effective study strategies

Instructors should incorporate instruction regarding critical engagement with AI (91). Strategies like spacing (i.e., spreading study sessions out over time), interleaving (i.e., mixing different topics in one study session), and retrieval practice (e.g., self-quizzing) have been consistently demonstrated to result in better long-term learning than passive review or cramming, which may help short-term recall but leads to quickly forgetting the material (5, 92, 93). However, many students use AI for instant explanations and answers that can create an illusion of understanding (94), much like rereading notes or highlighting without thinking through the material. This kind of passive use is reflected in the types of prompts students enter into the AI tool (88, Table 2) and does not activate the cognitive effort needed for actual learning (95). Therefore, we recommend that instructors dedicate a small amount of class time to facilitate the development of procedural knowledge regarding the use of AI to prepare for exams.

AI-produced practice questions are comparable in quality to instructor-created ones, particularly in terms of difficulty and conceptual focus, allowing students easy and fast access to self-testing (96). In using AI-generated tools, students may input their class notes and ask them to generate a summary or flashcards. Instructors can support student use of generative AI in this way by showing students how to create prompts that make AI give students tasks that require active engagement with the content. For example,

TABLE 2 Exemplar AI prompts that discourage or encourage the use of effective study behaviors

Ineffective prompt	Why is this ineffective?	Effective prompt	Why is this effective?
"Summarize Chapter 12 from my textbook and make a study guide for me."	• Generic outline creation • Encourages rereading • Promotes fluency illusion	"Using only the material in my Unit 2 notes, generate 10 mixed difficulty practice questions. Give me one question at a time. Do not show the answer until I attempt a response. After each question, ask me to explain why the correct answer is correct and why the distractors are wrong."	• Forces retrieval practice • Encourages self-explanation
"Make 50 flashcards for everything in Chapter 7."	• Produces flashcards that students scan or reread vs use for self-quizzing	"Using my notes, create 15 flashcards. Show me one card at a time and hide the answer until I try to recall it. If I get it wrong, send it back into the rotation."	• Encourages retrieval • Recycling missed cards supports spaced repetition
"Read my notes and tell me if I understand the material."	• Outsources metacognitive regulation to AI • Reinforces fluency illusion	"Based on my notes, generate a short quiz that tests whether I understand these concepts. Only reveal answers after I commit to a response."	• Encourages self-testing to allow students to evaluate their learning accurately
"Explain glycolysis to me in simple terms so I can remember it."	• Reinforces fluency illusion • Does not activate retrieval or elaboration • Encourages passive rereading	"Ask me to explain each step of glycolysis in my own words. After I respond, give me feedback on accuracy and ask one follow-up question to deepen my explanation."	• Encourages self-explanation • AI provides feedback to help students detect misconceptions

instead of using the prompt "Explain the steps of the Krebs cycle," which would quickly generate a summary for students to use, instructors could encourage students to use the prompt "Generate 10 mixed topic practice questions on cellular respiration, glycolysis, the Krebs cycle, and the electron transport chain. Give me one question at a time, and do not show the answers until I have tried each one." Similarly, students could be taught how to use prompts that create example problems where the AI gives partial solutions, and the student must finish them. This approach forces students to use AI in a way that encourages them to use effective, rather than ineffective, study strategies (97). Furthermore, to encourage the use of space and interleaving, instructors can demonstrate how to prompt AI to build a study schedule using the course syllabus. We provide additional examples of good AI prompts in Table 2.

Some instructors may advocate the use of AI for studying but are uncomfortable using AI due to copyright or intellectual property concerns arising from students uploading course materials. To address this, instructors should consider developing a course-specific AI tool that is aligned with the objectives and scope of the class. Kestin et al. (98) designed an instructor-designed AI tutor that leveraged pedagogical best practices (e.g., active learning and metacognitive evaluation) for use in an undergraduate physics course. Students who studied with the AI tutor outperformed students who used traditional active learning methods on course exams and reported increased engagement with the course content. Similarly, instructors could also design their own AI tool that prompts students to enter a concept they find challenging. The tool could then guide students through the practice questions or quizzes developed by the instructor. Additionally, the tool can provide instructor-generated explanations for course concepts. This approach would prevent hallucinations as the AI would be trained by the professor on the specific course. Furthermore, the AI could also be programmed to encourage the development of metacognitive regulation by providing prompts that ask students to rate their confidence following class assignments (e.g., judgments of learning [JOLs]) or identify areas of confusion.

Instructors are also encouraged to provide students with examples of AI-generated study materials. These examples are critical because they allow students to identify the features of a good study tool, and they provide students with opportunities for comparison (59). Finally, given that AI-generated output is prone to hallucinations,

instructors should demonstrate to students how to confirm that the information that is presented in the AI tool is consistent with what is present in their students' notes, course textbook, and other related course materials (99).

Recommendation 3: develop conditional knowledge of AI use for studying

To further deepen student metacognitive knowledge of AI-supported study tools, instructors should provide students with information regarding when to use these tools and why (i.e., conditional metacognitive knowledge) (45, 100). Student decisions about what study strategies to use are not only informed by the courses students are enrolled in but are also influenced by their exam expectations (2, 101–105). For example, when students perceive that their course has a heavy, fast-paced workload (102, 106), they rely on ineffective strategies (e.g., rereading their notes) to prepare for their exams. Similarly, if students believe that their exams will be multiple choice and require low cognitive effort (e.g., memorization), they will tailor their study approach to align with this expectation and primarily use ineffective strategies to prepare (10, 102, 105, 107, 108). This approach may further be reflected in how students engage with generative AI to study. If a student perceives that an exam in their biology course requires them to memorize, they will use AI in a manner that reflects this (e.g., using AI to generate vocabulary lists and summarize multiple lectures into one document). To counter this, instructors should inform students that AI should be used to facilitate active engagement with the course material while they study. Following this, instructors could ask students to generate a list of all the ways AI can be leveraged for active engagement.

Recommendation 4: encourage metacognitive regulation

In addition to providing knowledge regarding the benefits and limitations of generative AI, we recommend that instructors provide students with the skills necessary to actively monitor their learning processes (i.e., engage in metacognitive regulation) (91, 109). The use of generative AI may influence student metacognitive JOLs, where students rate their ability to recall the information that they just learned at a future time point (110). JOLs are influenced by the conditions available during the student's study period (e.g., having notes available or access to online resources) and can inform student decisions to continue or terminate the study (111). Using generative AI to study may promote "fluency illusion," where students interpret the ease of processing as a sign of effective learning and assume that they understand course content more than they do (i.e., overconfidence) (111–113). Indeed, previous studies have demonstrated that the use of basic online learning tools (e.g., texts presented online) during study time has resulted in students overestimating their understanding of the content that they have just learned, reduced student engagement, and limited metacognitive regulation (114). Further compounding the fluency illusion, generative AI can generate incorrect content due to delays in updating the platform (115), resulting in student acceptance of incorrect content, reinforcement of cognitive biases, and negative effects on academic performance. Collectively, this suggests that while using generative AI to study, students need guidance on assessing the information generated by these tools and actively monitoring their own learning (59, 109). Therefore, in order to develop students' evaluation skills related to AI-generated content, we recommend that instructors design assignments that require students to critically evaluate responses that are created by AI. For example, an instructor could ask AI to generate a quiz about the principles of mitosis. Once the quiz has been generated, students can then use their notes, textbooks, and other course materials to determine the veracity of the material presented. To encourage students to engage in metacognitive regulation, we recommend that instructors follow the steps outlined by Stanton and colleagues (45). Specifically, students should be provided with prompts to engage in metacognitive regulation. For example, following an exam, students could complete a post-exam assignment where they are asked to describe how they used AI to study and explain the effectiveness of this approach. Alternatively, students could provide answers to the questions provided in Box 1.

BOX 1. METACOGNITIVE REGULATION QUESTIONS THAT STUDENTS SHOULD ASK WHEN USING AI-GENERATED STUDY TOOLS

1. Is the AI-generated content consistent with what I learned in class?
2. Can I explain the reasoning behind the answer without looking?
3. Does the explanation match the level of detail my exam requires?
4. What misconceptions might this output contain? Did this study tool help me perform better on assessments? How?

Summary

The goal of this essay was to provide instructors with recommendations to help foster deep metacognitive knowledge of generative AI use and effective studying in students. Our recommendations were informed by current knowledge regarding AI use in undergraduate education, study strategies that positively influence academic performance, and best practices for developing student metacognition. Given that explicit instruction regarding these aspects has been consistently demonstrated to be beneficial for student academic outcomes, instructors are in a unique position to train and model to students how to study while also responsibly and effectively leveraging generative AI. We hope that the recommendations presented here will be used as a resource, along with co-curricular study skills courses and course materials, to develop student metacognitive knowledge regarding AI-supported study behaviors.

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